

U.S. Distribution Center Location Strategy for Consumer Electronics

A data-driven supply chain case study using Census, CBP, ACS, and FAF data

Final recommendation: Chicago
Strongest alternative: Dallas-Fort Worth

Prepared as a portfolio case study for supply chain network planning, distribution strategy, and data-driven location analysis.

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EXECUTIVE SUMMARY

This case study evaluates where a consumer electronics distributor should open a new U.S. distribution center. The analysis compares Chicago, Dallas-Fort Worth, Atlanta, Phoenix, and Columbus using a weighted scorecard built from public datasets. The goal was not to find the cheapest warehouse location, but to identify the strongest overall distribution market based on demand, freight access, logistics infrastructure, labor availability, growth potential, and cost environment.

The final model ranked Chicago first with a score of 77.2 out of 100. Dallas-Fort Worth ranked second with a score of 75.7, making it a very close alternative. Chicago won because it performed extremely well in logistics infrastructure, market demand, freight access, and labor availability. Dallas-Fort Worth was stronger in growth potential and cost environment, which makes it a strong alternative if the company is prioritizing long-term growth and lower operating cost.

Rank	City	Final Score /100	Main Interpretation
1	Chicago	77.2	Best overall existing logistics market; strongest infrastructure and demand profile.
2	Dallas-Fort Worth	75.7	Best alternative; strong growth, demand, labor, and cost balance.
3	Atlanta	57.9	Useful Southeast hub, but weaker than Chicago and Dallas in final model.
4	Phoenix	40.1	Strong growth, but lower current logistics scale and labor depth.
5	Columbus	28.7	Cost-friendly, but smaller demand and infrastructure base.

FINAL RECOMMENDATION

The recommended location is Chicago. The recommendation is based on the city receiving the highest total score, not because it was perfect in every category. Chicago performed poorly in growth potential and cost environment, but its advantages in logistics infrastructure, demand, freight activity, and labor availability outweighed those weaknesses in the final model.

Dallas-Fort Worth should be considered the strongest alternative. The difference between Chicago and Dallas-Fort Worth was only 1.5 points on the 100-point scale, so the decision is close. If the company prioritizes future market growth, cost balance, and Sun Belt expansion, Dallas-Fort Worth could become the preferred option.

BACKGROUND CONTEXT

Consumer electronics distribution is built around speed, accuracy, and network reach. Products such as phones, tablets, laptops, chargers, headphones, speakers, and accessories are usually high-value and relatively lightweight. That creates a different distribution challenge than industries that move heavy bulk goods. A consumer electronics company needs access to high-demand consumer markets, reliable truck movement, strong parcel and regional delivery options, warehousing capacity, return flows, and enough labor to support receiving, picking, packing, shipping, and reverse logistics.

A distribution center location affects transportation cost, delivery time, customer service, labor availability, returns processing, and long-term scalability. A poor location can increase freight distance, slow delivery, and create labor or capacity problems. A strong location gives the company better access to customers, carriers, warehouses, and supply chain support services.

This project uses public data to simulate the kind of location decision a supply chain analyst might support. Because internal company data such as exact customer addresses, order volume, supplier locations, shipment frequency, and transportation rates was not available, the model uses public metro-level data as proxies for demand, freight strength, infrastructure, labor, growth, and operating cost.

PROCEDURE OVERVIEW: IDEA, PLANNING, PROCESS, EXECUTION, AND REFLECTION

IDEA

The original idea was to create a high-level supply chain project using Census data. Instead of building a generic dashboard, the project was framed around a real business decision: selecting the best U.S. distribution center location for a consumer electronics distributor. This made the project more practical because it connects public data to a decision that supply chain teams actually make.

PLANNING

The planning stage focused on defining the business problem, choosing comparison markets, selecting data sources, and building a scoring model before collecting data. This step mattered because it prevented the project from becoming a random collection of numbers. Each dataset had to support a specific decision factor.

Decision Factor	Why It Was Included
Market Demand	Shows where customer and business demand is strongest.
Freight Access	Shows how easily goods can move in and out of the region.
Logistics Infrastructure	Shows whether the region already has warehousing, trucking, and distribution support.
Labor Availability	Shows whether the region has enough workers and supply chain labor depth.
Growth Potential	Shows whether the market is expanding over time.
Cost Environment	Provides a simple labor-cost proxy for operating expense.

PROCESS

The process used separate spreadsheet tabs for each category. Each category sheet collected raw data, converted each metric into a normalized 1-to-10 score, and then calculated a final category score using category-specific weights. A final scorecard then pulled the category scores together and applied the overall model weights.

EXECUTION

The execution stage included downloading or using metro-level data from the American Community Survey, the 2023 County Business Patterns MSA file, and the Freight Analysis Framework. The results were organized into a

spreadsheet model and then visualized in Tableau. The Tableau dashboard focused on final ranking, scoring weights, category comparisons, and the tradeoff between Chicago and Dallas-Fort Worth.

REFLECTION

The biggest lesson from the project is that location decisions are not one-dimensional. A city can be strong in logistics infrastructure but weak in cost or growth. A city can be fast-growing but not yet have the same logistics ecosystem as an established freight hub. The project also showed the importance of checking data definitions carefully. For example, the Freight Analysis Framework splits Chicago into multiple freight regions, so the Chicago IL and IN parts had to be combined to avoid understating Chicago freight activity.

DATA SOURCES AND HOW THEY WERE USED

Three main public data sources were used: the American Community Survey, County Business Patterns, and the Freight Analysis Framework. The sources were selected because they provide consistent public data at regional or industry levels that can support a high-level supply chain decision.

Source	Used For	Why It Mattered
American Community Survey (ACS)	Population, households, median income, commute time, and population growth	Supported demand, labor, and growth analysis.
County Business Patterns (CBP), 2023 MSA File	Establishments, employment, and annual payroll by industry and metro area	Supported market demand, logistics infrastructure, labor availability, and cost environment.
Freight Analysis Framework (FAF)	Freight value by origin, destination, mode, and region	Supported freight access analysis.

CBP was especially important because it includes number of establishments, employment, first quarter payroll, and annual payroll by industry and geography. The project used the MSA file because the analysis compared metro markets rather than city limits or ZIP codes.

FAF was used because it estimates freight flows by origin, destination, mode, commodity, value, and weight. For this project, freight value was more useful than tonnage because consumer electronics are high-value and relatively lightweight compared with bulk commodities.

GEOGRAPHY CHOICE

The project used metro areas and closest FAF freight regions instead of city-only data. This matters because a distribution center does not serve only the legal city boundary. A warehouse in the Chicago, Dallas-Fort Worth, Atlanta, Phoenix, or Columbus market serves the broader region.

Project Market	CBSA Code	FAF Region Used
Dallas-Fort Worth	19100	484-Dallas-Fort Worth TX-OK (TX Part)
Atlanta	12060	131-Atlanta GA
Chicago	16980	171-Chicago IL-IN-WI (IL Part) + 181-Chicago IL-IN-WI (IN Part)
Phoenix	38060	041-Phoenix AZ
Columbus	18140	393-Columbus OH

IMPORTANT DATA LIMITATION

FAF regions do not perfectly match Census CBSA metro areas. The project uses the closest FAF metropolitan freight regions as proxies. For Chicago, two Chicago-labeled FAF parts were combined because using only one part would understate the region. This is acceptable for a high-level location strategy, but it should not be interpreted as exact facility-level freight volume.

SCORING METHODOLOGY

The model used a weighted scorecard. Every category was scored from 1 to 10, then each category was multiplied by its overall model weight. The final weighted score was multiplied by 10 to produce a final score out of 100.

Category	Overall Weight
Market Demand	25%
Freight Access	25%
Logistics Infrastructure	20%
Labor Availability	15%
Growth Potential	10%
Cost Environment	5%

Final weighted score formula:

Final Weighted Score = (Market Demand x 0.25) + (Freight Access x 0.25) + (Logistics Infrastructure x 0.20) + (Labor Availability x 0.15) + (Growth Potential x 0.10) + (Cost Environment x 0.05)

Final Score /100 = Final Weighted Score x 10

NORMALIZATION FORMULA

Most raw metrics were converted into a 1-to-10 score using min-max normalization. The city with the highest value received a 10, the city with the lowest value received a 1, and the other cities were scored proportionally between them.

Standard score formula:

Score = 1 + 9 x ((City Value - Minimum Value) / (Maximum Value - Minimum Value))

For metrics where lower is better, such as commute time or average warehouse payroll per employee, the formula was reversed.

Reverse score formula:

Reverse Score = 1 + 9 x ((Maximum Value - City Value) / (Maximum Value - Minimum Value))

MARKET DEMAND ANALYSIS

PURPOSE

Market Demand measured the potential customer base for consumer electronics. A distribution center should be close to regions with large populations, large household bases, retail activity, electronics-related retail activity, and purchasing power.

Metric		Weight	How It Was Used			
Metro Population		30%	Larger population suggests a larger consumer electronics market.			
Households		20%	More households means more potential consumer electronics buyers.			
Median Household Income		10%	Used as a purchasing power indicator.			
Retail Businesses		20%	Used as a B2B distribution demand indicator.			
Electronics/Appliance Retailers		20%	Used as an electronics-specific retail demand indicator.			
City	Population	Households	Median Income	Retail Businesses	Electronics/Appliance Retailers	Demand Score
Dallas-Fort Worth	8,344,032	3,012,855	\$92,733	21,680	477	8.70
Atlanta	6,409,047	2,390,346	\$92,344	18,625	338	6.63
Chicago	9,406,924	3,690,402	\$90,770	25,308	507	9.82
Phoenix	5,186,958	1,967,378	\$90,133	11,620	237	4.59
Columbus	2,225,377	880,121	\$82,938	5,594	111	1.00

Chicago scored highest in Market Demand because it had the largest population, the largest number of households, the highest retail establishment count, and the highest electronics/appliance retailer count among the five markets. Dallas-Fort Worth ranked second and was also strong because of its large population, strong income, and large retail base.

FREIGHT ACCESS ANALYSIS

PURPOSE

Freight Access measured how well each region could support the movement of goods. For a consumer electronics distributor, freight value is more useful than tonnage because electronics are high-value and relatively lightweight. The model therefore focused on total freight value and truck share.

Metric		Weight	How It Was Calculated
Total Freight Value		70%	Outbound freight value + inbound freight value from FAF.
Truck Share		30%	Total truck freight value divided by total freight value.

Freight formulas used:

Total Freight Value = Outbound Freight Value + Inbound Freight Value

Total Truck Freight Value = Truck Outbound Freight Value + Truck Inbound Freight Value

Truck Share = Total Truck Freight Value / Total Freight Value

City	Outbound Freight Value	Inbound Freight Value	Total Freight Value	Truck Share	Freight Score
Dallas-Fort Worth	\$589,888.7M	\$535,522.9M	\$1,125,411.6M	77.8%	5.67
Atlanta	\$375,514.8M	\$382,154.4M	\$757,669.2M	77.0%	3.55
Chicago	\$844,229.1M	\$675,663.0M	\$1,519,892.1M	76.7%	7.30
Phoenix	\$106,548.5M	\$175,449.0M	\$281,997.5M	85.0%	3.70
Columbus	\$176,436.9M	\$205,436.5M	\$381,873.4M	82.8%	3.49

Chicago ranked first in Freight Access because it had the strongest total freight value after combining both Chicago FAF parts. Dallas-Fort Worth ranked second. Phoenix had the highest truck share, but it did not have enough total freight value to beat Chicago or Dallas-Fort Worth. This is why total freight value received a higher weight than truck share.

Data quality note: Freight values should be reviewed before final publication if any truck-only value appears larger than the corresponding all-mode value. Since all-mode freight includes truck, truck freight should not exceed the all-mode total for the same origin or destination setup. The project used the final category scores from the completed scorecard, but detailed FAF values should be rechecked if the spreadsheet is updated.

LOGISTICS INFRASTRUCTURE ANALYSIS

Logistics Infrastructure measured whether each region already has the business ecosystem needed to support distribution center operations. This category is different from Freight Access. Freight Access measures movement of goods, while Logistics Infrastructure measures the presence of warehouses, trucking firms, wholesalers, transportation support firms, and electronics wholesalers.

Metric	NAICS Code		Weight			
Warehousing Businesses	493///		30%			
Trucking Businesses	484///		25%			
Wholesale Trade Businesses	42----		20%			
Transportation Support Businesses	488///		15%			
Electronics Wholesalers	4236//		10%			
City	Warehousing	Trucking	Wholesale	Transport Support	Electronics Wholesalers	Infrastructure Score
Dallas-Fort Worth	703	3,420	9,324	1,246	810	6.82
Atlanta	602	3,346	8,164	1,124	600	5.67
Chicago	727	12,968	12,261	2,218	865	10.00
Phoenix	270	1,564	4,815	579	383	2.14
Columbus	319	989	1,938	253	128	1.29

Chicago dominated this category because it had the strongest concentration of logistics-related establishments. It had the highest trucking count, wholesale trade count, transportation support count, electronics wholesaler count, and one of the strongest warehouse establishment counts. This category was one of the main reasons Chicago ranked first overall.

LABOR AVAILABILITY ANALYSIS

Labor Availability measured the depth of the workforce that could support a distribution center. A warehouse needs workers for receiving, putaway, picking, packing, inventory control, outbound shipping, returns, and supervision. The model used total employment, transportation and warehousing employment, warehousing employment, and mean commute time.

Metric	Weight			Scoring Direction	
Total Employment	25%			Higher is better	
Transportation and Warehousing Employment	35%			Higher is better	
Warehousing Employment	25%			Higher is better	
Mean Commute Time	15%			Lower is better	
City	Total Employment	Transportation/Warehousing Employment	Warehousing Employment	Mean Commute Time	Labor Score
Dallas-Fort Worth	3,594,749	264,835	71,316	28.8	8.44
Atlanta	2,692,513	221,772	67,772	32.4	6.36
Chicago	4,285,971	284,408	65,833	30.9	8.63
Phoenix	2,080,244	108,819	24,134	27.6	2.78
Columbus	985,064	93,843	53,258	24.1	3.74

Chicago ranked first in Labor Availability because it had the strongest total employment and transportation/warehousing employment base. Dallas-Fort Worth was very close because it had the strongest warehousing employment count and a large overall labor market. Columbus had a better commute time but a smaller overall labor base, so it did not rank near the top.

GROWTH POTENTIAL ANALYSIS

Growth Potential measured whether each market was expanding. Since this category was only 10% of the final model, the analysis focused on population growth rather than overcomplicating the model with less consistent growth indicators.

Metric		Weight		How It Was Used	
Population Growth %		70%		Measures how fast the metro area is growing.	
Population Added		30%		Measures the absolute number of people added.	
City	2020 Population	2024 Population	Population Added	Growth %	Growth Score
Dallas-Fort Worth	7,637,387	8,344,032	706,645	9.25%	10.00
Atlanta	6,089,815	6,409,047	319,232	5.24%	6.60
Chicago	9,600,822	9,406,924	-193,898	-2.02%	1.00
Phoenix	4,845,832	5,186,958	341,126	7.04%	7.67
Columbus	2,138,926	2,225,377	86,451	4.04%	5.23

Dallas-Fort Worth won Growth Potential because it added the most people and had the highest population growth percentage among the five markets. Phoenix also performed strongly. Chicago scored lowest because its metro population declined over the comparison period.

COST ENVIRONMENT ANALYSIS

Cost Environment was included as a small but important factor. A complete cost model would include warehouse rent, utilities, taxes, insurance, incentives, and transportation rates. Because consistent real estate cost data was not used across all five metros, this project used average warehousing payroll per employee as a labor-cost proxy.

Metric	Weight	Scoring Direction
Average Warehousing Payroll per Employee	100%	Lower is better

The calculation was:

Average Warehousing Payroll per Employee = (Warehousing Annual Payroll in \$1,000s x 1,000) / Warehousing Employment

City	Warehousing Employment	Warehousing Annual Payroll (\$1,000s)	Average Warehouse Payroll	Cost Score
Dallas-Fort Worth	71,316	3,520,498	\$49,364.77	6.85
Atlanta	67,772	3,138,984	\$46,316.83	10.00
Chicago	65,833	3,621,647	\$55,012.64	1.00
Phoenix	24,134	1,199,950	\$49,720.31	6.48
Columbus	53,258	2,566,515	\$48,190.22	8.06

Atlanta scored highest in Cost Environment because it had the lowest average warehousing payroll per employee. Chicago scored lowest because it had the highest average warehousing payroll per employee. This does not mean Chicago is a bad location, but it does show a cost tradeoff.

FINAL SCORECARD AND RECOMMENDATION

The final scorecard combined the six category scores using the overall category weights. The purpose was to create a clear decision model that could compare very different location factors on the same 100-point scale.

City	Market Demand	Freight Access	Logistics Infrastructure	Labor Availability	Growth Potential	Cost Environment	Final Score /100	Rank
Dallas-Fort Worth	8.70	5.67	6.82	8.44	10.00	6.85	75.7	2
Atlanta	6.63	3.55	5.67	6.36	6.60	10.00	57.9	3
Chicago	9.82	7.30	10.00	8.63	1.00	1.00	77.2	1
Phoenix	4.59	3.70	2.14	2.78	7.67	6.48	40.1	4
Columbus	1.00	3.49	1.29	3.74	5.23	8.06	28.7	5

WHY CHICAGO WON

Chicago won because it had the strongest combination of current operating advantages. It ranked first in Market Demand, Logistics Infrastructure, and Labor Availability, and it also performed well in Freight Access. Those categories made up 85% of the final model weight. Even though Chicago scored poorly in Growth Potential and Cost Environment, those two categories made up only 15% of the final model.

WHY DALLAS-FORT WORTH WAS CLOSE

Dallas-Fort Worth was only 1.5 points behind Chicago. Dallas-Fort Worth performed strongly across demand, labor, growth, and cost. It also had a strong logistics ecosystem, although not as large as Chicago. Its biggest weakness in the final model was Freight Access, where Chicago had stronger total freight value.

BUSINESS RECOMMENDATION

The recommended location is Chicago if the company wants the strongest established distribution market. Chicago offers a large demand base, deep logistics infrastructure, strong labor availability, and high freight activity. It is the best choice for a company that prioritizes immediate operational strength and access to a mature national logistics hub.

Dallas-Fort Worth should be treated as the strongest alternative. If the company prioritizes long-term population growth, lower operating cost, and expansion into a fast-growing Sun Belt market, Dallas-Fort Worth could be preferred despite ranking slightly behind Chicago in the final model.

DASHBOARD DEVELOPMENT

After building the spreadsheet model, the results were converted into Tableau-ready tabs. The dashboard was designed to tell the decision story visually rather than display every raw data point. The dashboard focused on four visuals: final city ranking, scoring model weights, category score heatmap, and Chicago vs Dallas-Fort Worth tradeoff comparison.

Dashboard Visual	Purpose
Final Distribution Center Location Ranking	Shows the final score and city ranking.
Scoring Model Weight Breakdown	Explains how the model was weighted.
Category Score Comparison by City	Shows each city's strengths and weaknesses across categories.
Chicago vs Dallas-Fort Worth Tradeoff	Explains why Chicago won but Dallas was close.

U.S. Distribution Center Location Strategy for Consumer Electronics

A data-driven comparison of Chicago, Dallas-Fort Worth, Atlanta, Phoenix, and Columbus

Key Insight: Chicago ranked first due to strong logistics infrastructure, market demand, freight access, and labor availability. Dallas-Fort Worth ranked close behind and is the best alternative because of stronger gr..

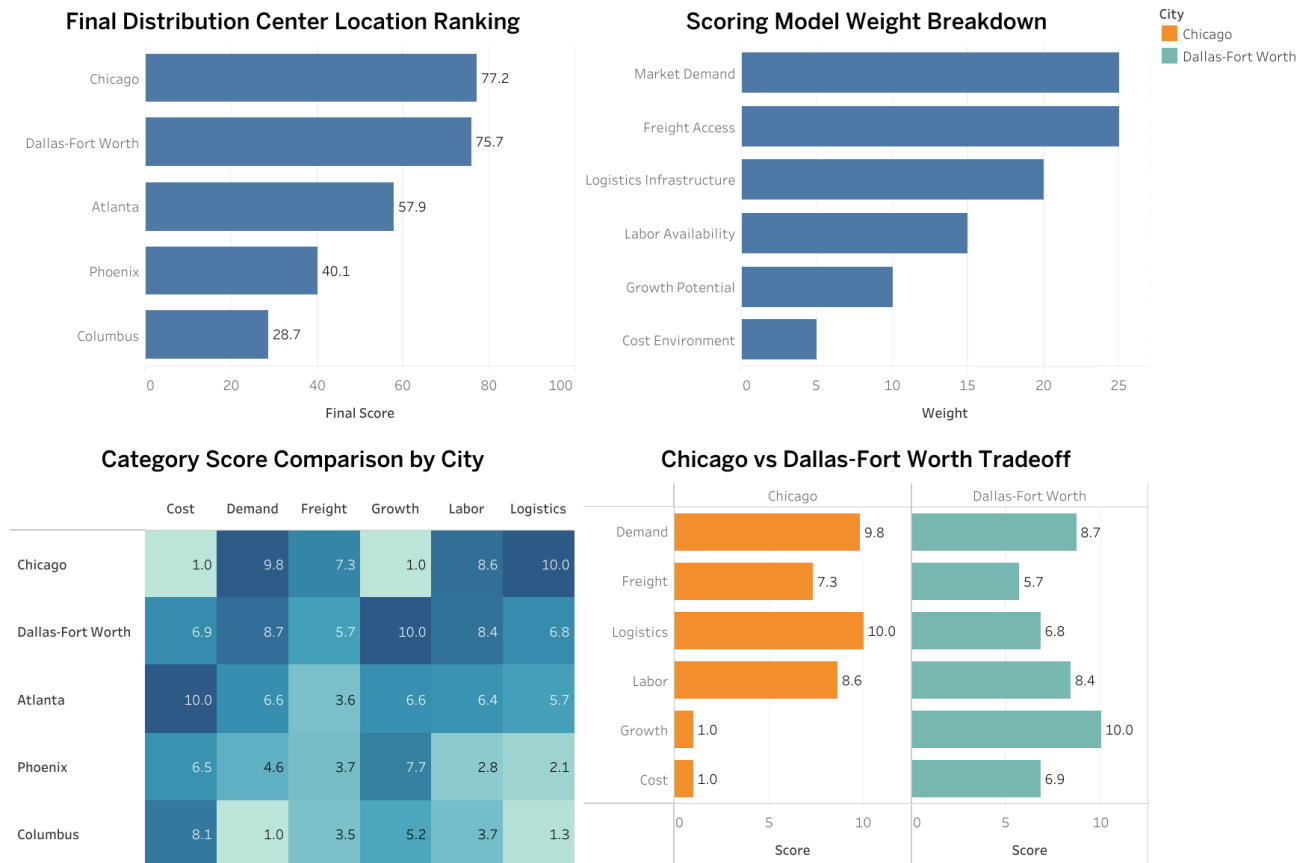


Figure 1. Tableau dashboard showing the final ranking, category weights, heatmap, and Chicago vs Dallas-Fort Worth tradeoff.

The dashboard reinforced the main conclusion. Chicago ranked first overall, but Dallas-Fort Worth was very close. The heatmap showed that Chicago was strong in demand, freight, logistics, and labor, while Dallas-Fort Worth was much stronger in growth and cost. This made the decision more realistic because the winner was not perfect in every category.

REFLECTION AND LIMITATIONS

WHAT WORKED WELL

- The project connected public data to a real supply chain decision instead of creating a generic data dashboard.
- The scoring model was structured before collecting data, so each dataset had a clear purpose.
- Using metro-level data made more sense than city-limit data because distribution centers serve regional markets.
- The final result produced a realistic tradeoff: Chicago was strongest today, while Dallas-Fort Worth was the strongest alternative for growth and cost.

LIMITATIONS

- The model does not include actual customer order locations, supplier locations, transportation rates, or facility-level costs.
- FAF freight regions do not perfectly match CBSA metro areas, so the freight analysis uses the closest available freight-region proxies.
- Cost Environment used average warehousing payroll per employee as a proxy. A full cost model would also include warehouse rent, taxes, utilities, insurance, incentives, and transportation costs.
- The model uses public data and therefore should be treated as a high-level strategy model, not a final site selection model.
- Freight values should be rechecked if the underlying FAF filtering changes, especially where a mode-specific value appears inconsistent with an all-mode value.

WHAT I WOULD IMPROVE NEXT

- Add warehouse rent data from a consistent commercial real estate source for all five metros.
- Add drive-time analysis from each city to major customer markets.
- Add parcel carrier hub locations and airport cargo indicators for consumer electronics fulfillment.
- Add sensitivity analysis to test whether Chicago still wins if growth or cost receive higher weights.
- Build a second model for a two-warehouse network to compare Chicago plus Dallas-Fort Worth against a single-location strategy.

The most important reflection is that the best location depends on the company's strategy. If the company wants the strongest existing logistics hub, Chicago is the best recommendation. If the company wants a lower-cost, high-growth alternative, Dallas-Fort Worth becomes very compelling. A good supply chain recommendation should explain the tradeoff rather than pretending the highest score is automatically perfect.

APPENDIX A: METRIC DEFINITIONS

Metric	Definition / Calculation
Metro Population	Population of the metro area from ACS/Census-based metro profile data.
Households	Number of households in the metro area.
Median Household Income	Median household income, used as a purchasing power proxy.
Retail Businesses	CBP establishment count for NAICS 44----.
Electronics/Appliance Retailers	CBP establishment count for NAICS 443/// in the downloaded file.
Outbound Freight Value	FAF value of freight shipped from the region to all destinations.
Inbound Freight Value	FAF value of freight shipped from all origins into the region.
Total Freight Value	Outbound Freight Value + Inbound Freight Value.
Total Truck Freight Value	Truck Outbound Freight Value + Truck Inbound Freight Value.
Truck Share	Total Truck Freight Value / Total Freight Value.
Warehousing Businesses	CBP establishment count for NAICS 493///.
Trucking Businesses	CBP establishment count for NAICS 484///.
Wholesale Trade Businesses	CBP establishment count for NAICS 42----.
Transportation Support Businesses	CBP establishment count for NAICS 488///.
Electronics Wholesalers	CBP establishment count for NAICS 4236//.
Average Warehouse Payroll	(Warehousing Annual Payroll in \$1,000s x 1,000) / Warehousing Employment.

APPENDIX B: SOURCES USED

The project used the following source types and pages:

- U.S. Census Bureau, County Business Patterns (CBP): <https://www.census.gov/programs-surveys/cbp.html>
- U.S. Census Bureau, 2023 County Business Patterns datasets: <https://www.census.gov/programs-surveys/cbp/data/datasets.html>
- U.S. Census Bureau, American Community Survey (ACS): <https://www.census.gov/programs-surveys/acs.html>
- U.S. Census Bureau, Commuting / Journey to Work data description: <https://www.census.gov/topics/employment/commuting.html>
- Bureau of Transportation Statistics, Freight Analysis Framework (FAF): <https://www.bts.gov/faf>
- Federal Highway Administration, Freight Analysis Framework information: https://ops.fhwa.dot.gov/freight/freight_analysis/faf/
- Census Reporter metro profiles were used as an accessible interface for ACS metro-level population, income, household, and commute values: <https://censusreporter.org/>

Note: Values from the 2023 CBP MSA file were taken from the uploaded Census data file and filtered by CBSA code and NAICS code. FAF values were taken from the FAF tool by selecting 2023, value, all commodities, relevant origin/destination freight region, and truck mode where applicable.